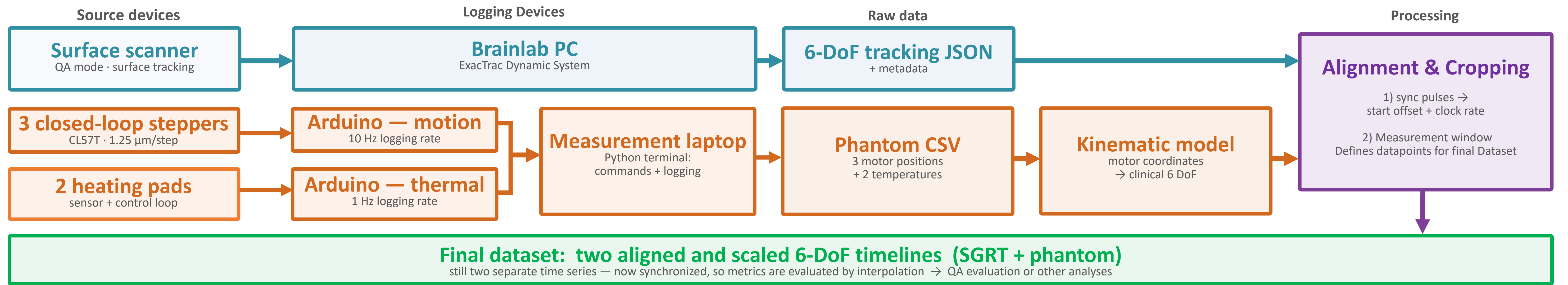


Two independent systems, one analyzable pair of timelines



A. Mechanical realization

- Open, off-the-shelf parts throughout: PLA 3D print, CNC closed-loop steppers with CL57T drivers, 20x20 mm profiles
- Two axes with 4 mm spindles are kinematically coupled — accurate without a robot, and the coupling produces the motion that can be used to fake a respiratory trace. A rotational Degree of Freedom (DoF) sits on top.

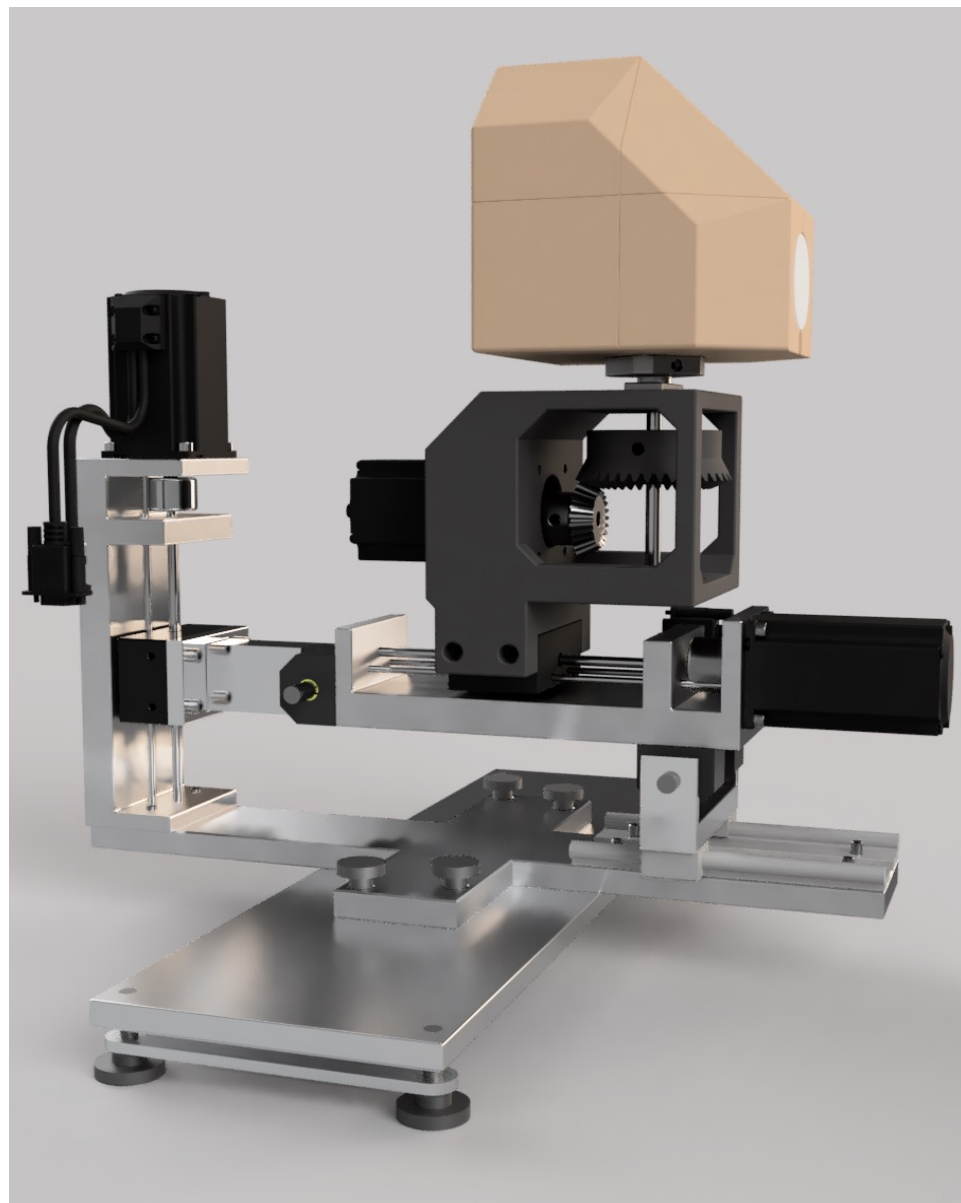


Fig. 1: CAD Rendering of initial Design

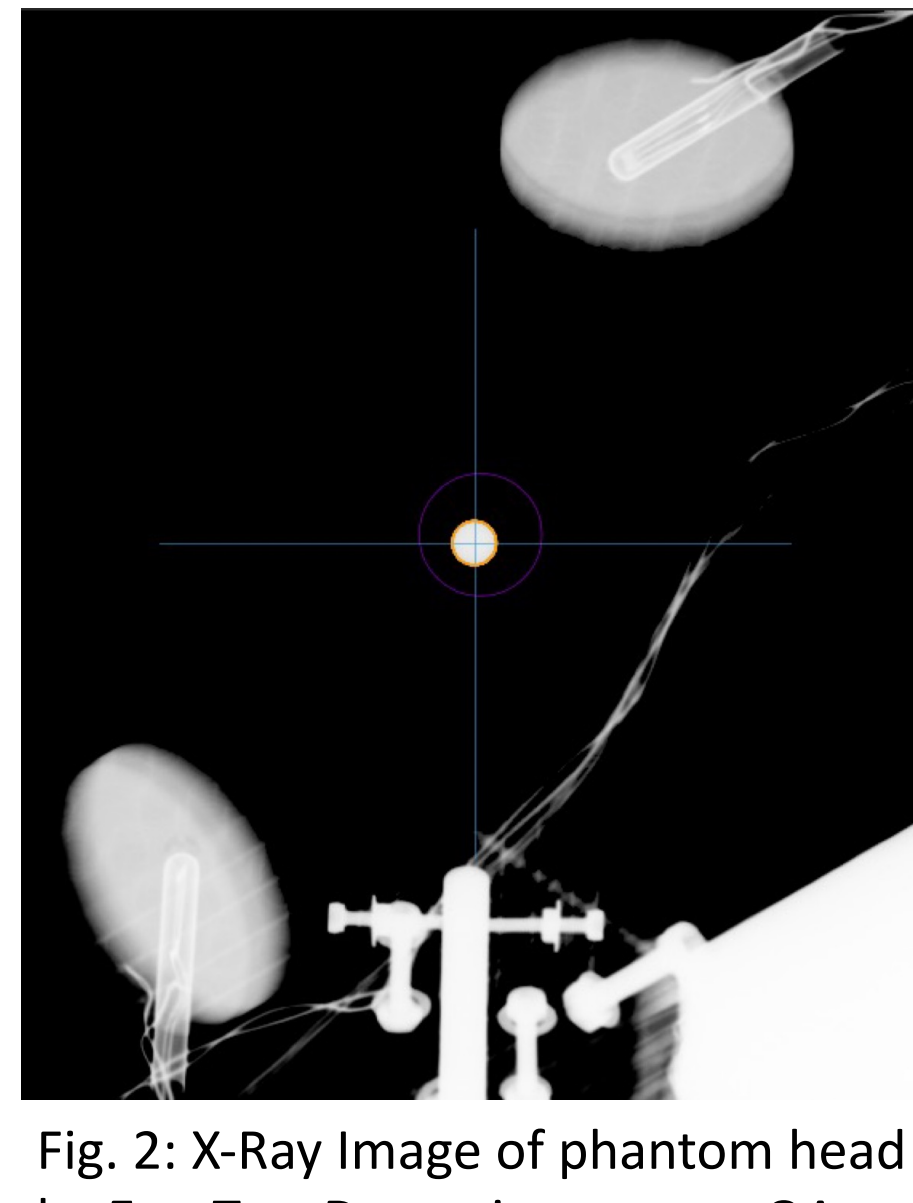


Fig. 2: X-Ray Image of phantom head by ExacTrac Dynamic system – QA sphere detection (SD) mode

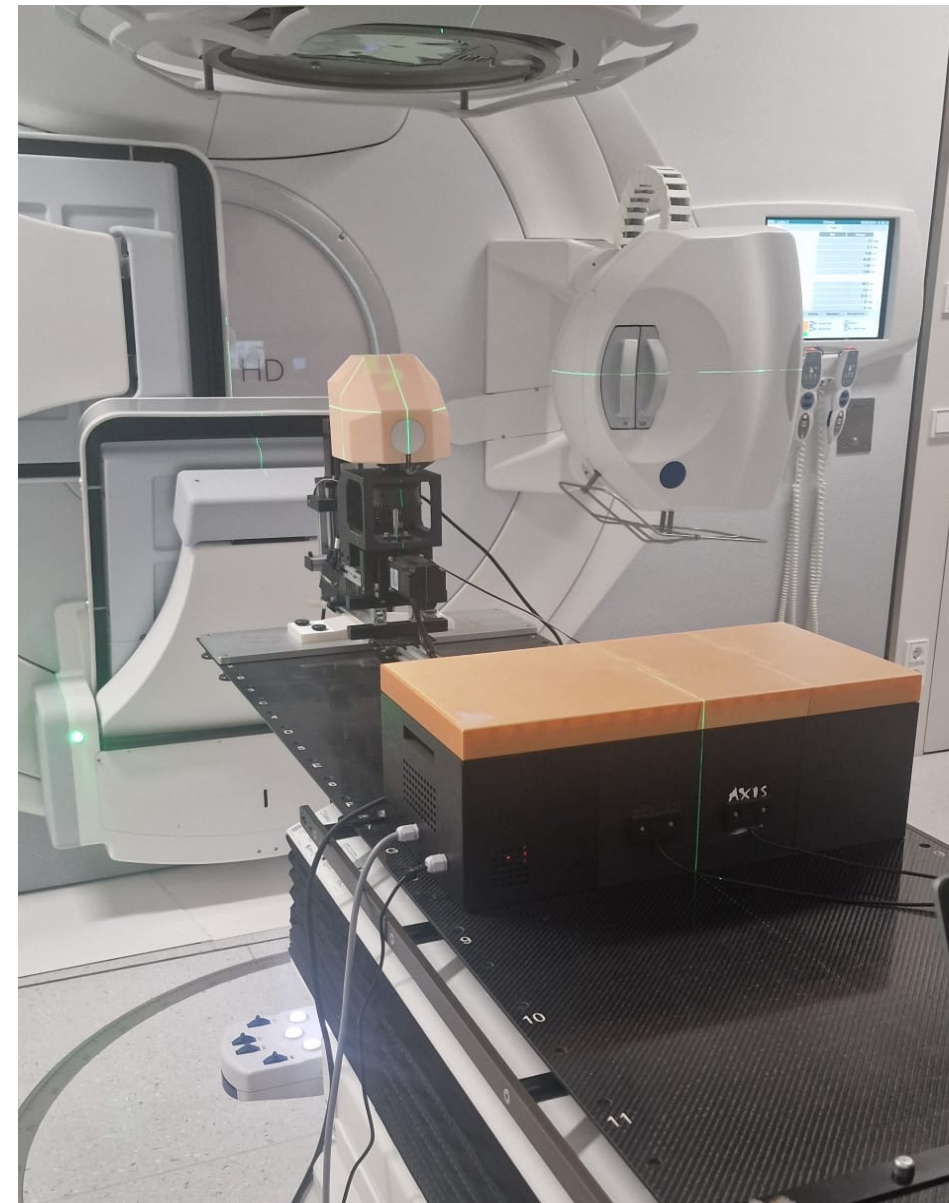


Fig. 3: Phantom - on treatment couch

B. Electronics & control

- Two separate Arduino Nanos: logging motion data at 10 Hz, thermal data at 1 Hz, simple PCBs for the sensors.
- Serial protocol to Python terminal with commands, operating modes and logging in one place
- Two heating elements: 5 W flexible heatpads sandwiched between 5 cm aluminum discs, featuring a DS18B20 temperature sensor, bonded with thermally conductive adhesive tape and surface finished with thermal coating

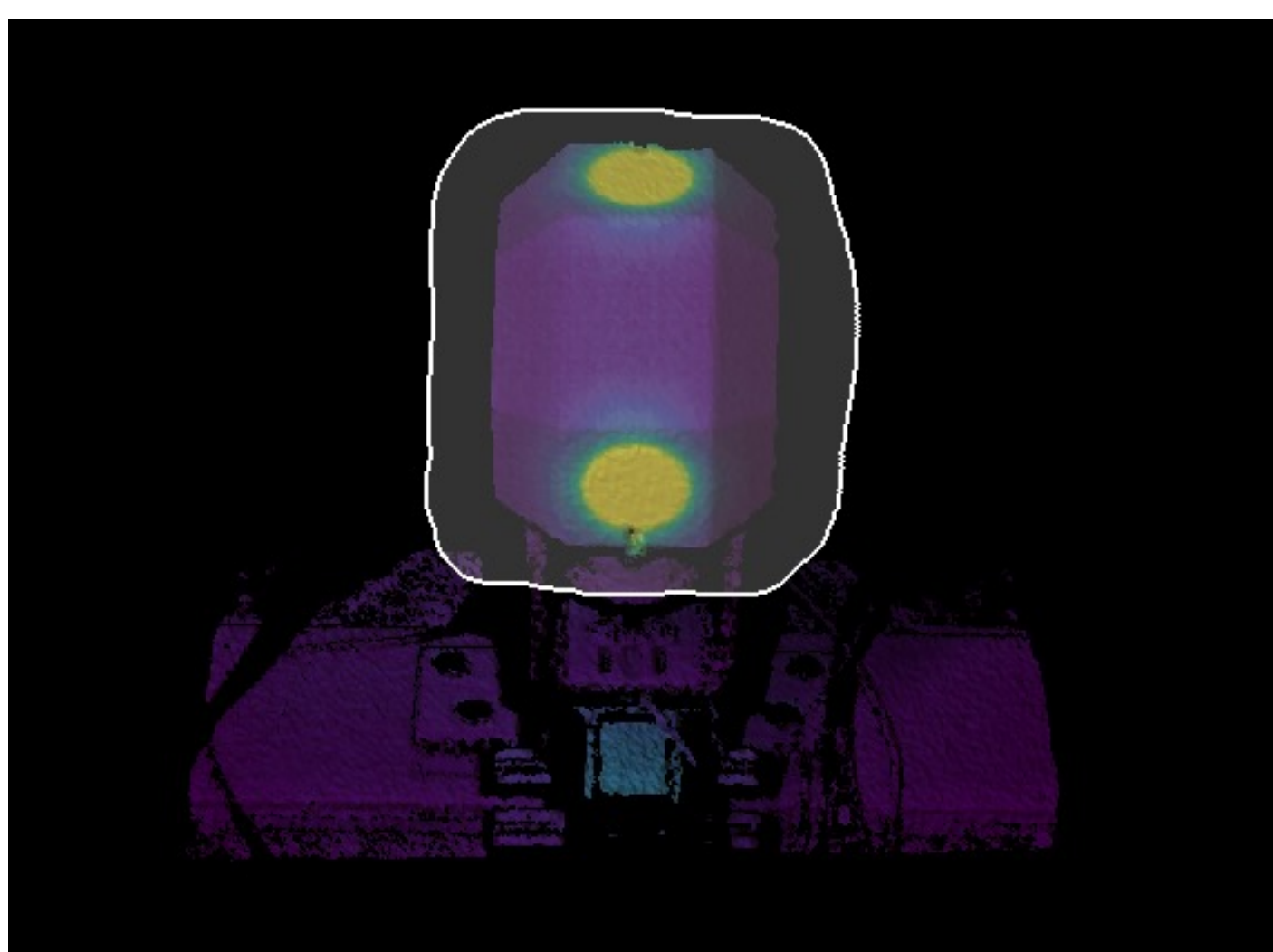


Fig. 4: image from ETD system – pads at 32°C

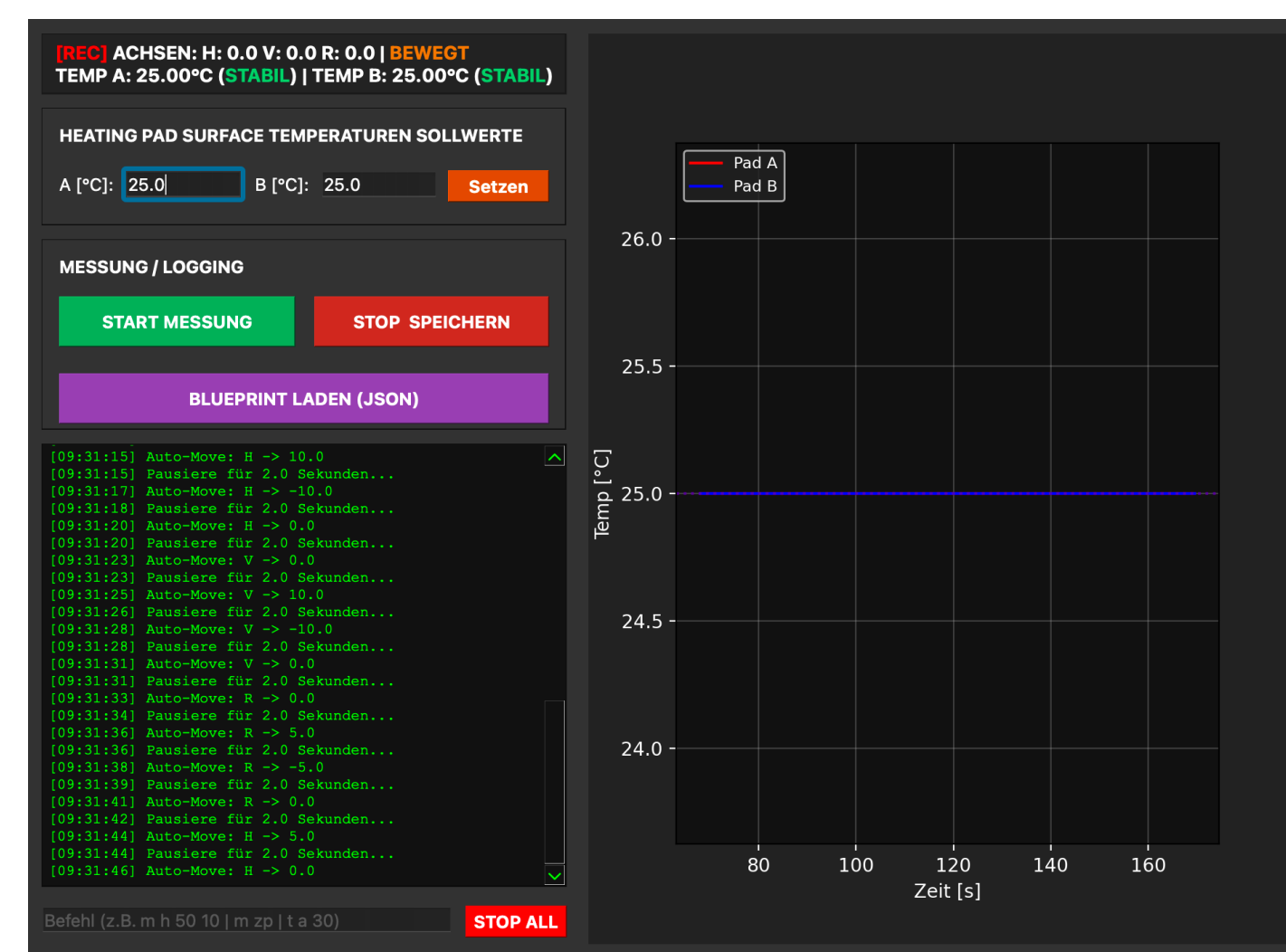


Fig. 5: Python terminal on measurement pc

E. Generating analyzable timelines

- Both systems log on their own free-running clock; each run is bracketed by sync pulses.
- Recovered per run: start offset and relative clock rate, then a fixed measurement window.

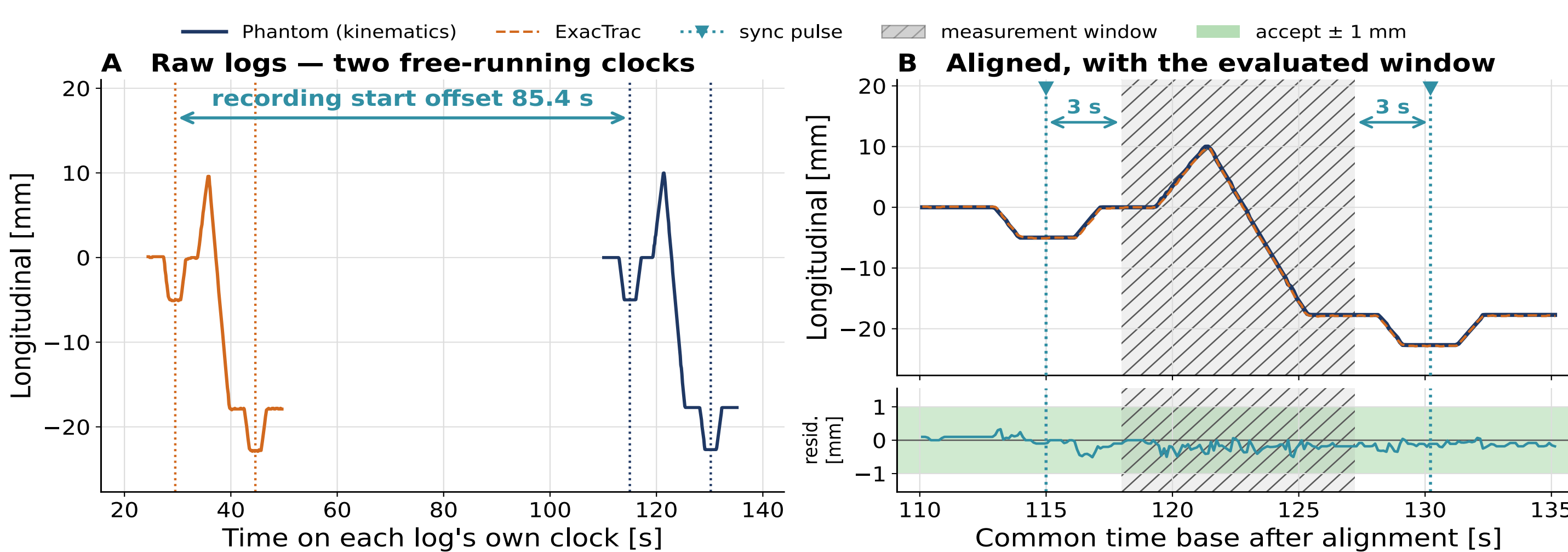


Fig. 6: process of timeline alignment (Data QA Deflection 1: A – Before, B – After with signal residuals)

C. Motion concept & workspace

- Three axes, two of them coupled: horizontal translation, and a vertical/pitch axis driven through a 3D-printed bevel gear – lateral (and roll) motion is realized by rotating the treatment couch to 270° and phantom to 90°
- The reachable workspace defines where pure translations exist and where motion is necessarily combined.

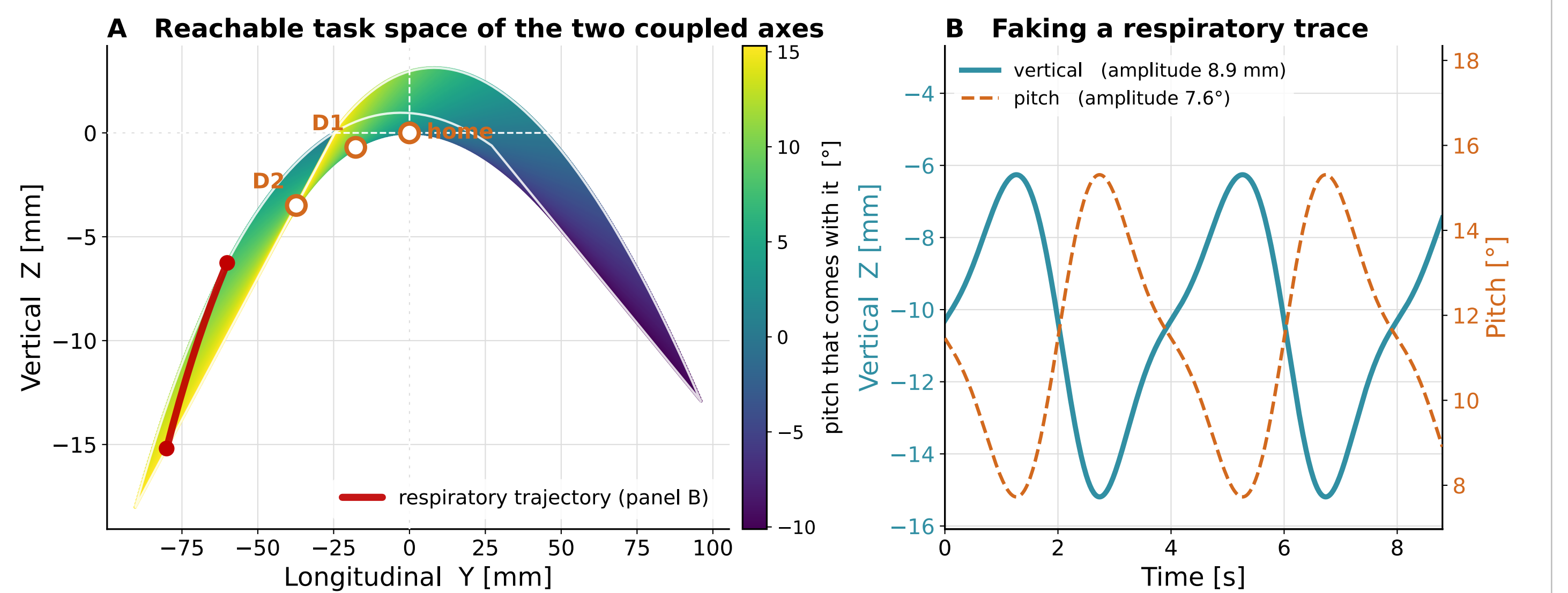


Fig. 7: (A) Simulated workspace of our phantom: Due to mechanical axis coupling, no pure vertical translation exists – vertical movement inherently causes pitch movement. D1, D2 are annual QA deflections (see QR code). (B) shows a hypothetical respiratory trajectory (with amplitude 8.9 mm vert., 7.6° pitch) constrained to 20 mm longitudinal travel.

D. Axis validation vs. sphere detection

- Linear axis: 4 mm lead with micro-stepping = 1.25 μm per step, monitored by the motor encoder - a commanded 10 mm longitudinal shift is reproduced to 0.0 mm by X-ray sphere detection
- Rotation: the sphere sits 1.1 mm off the rotation axis, producing an apparent shift up to 1.5 at 90°

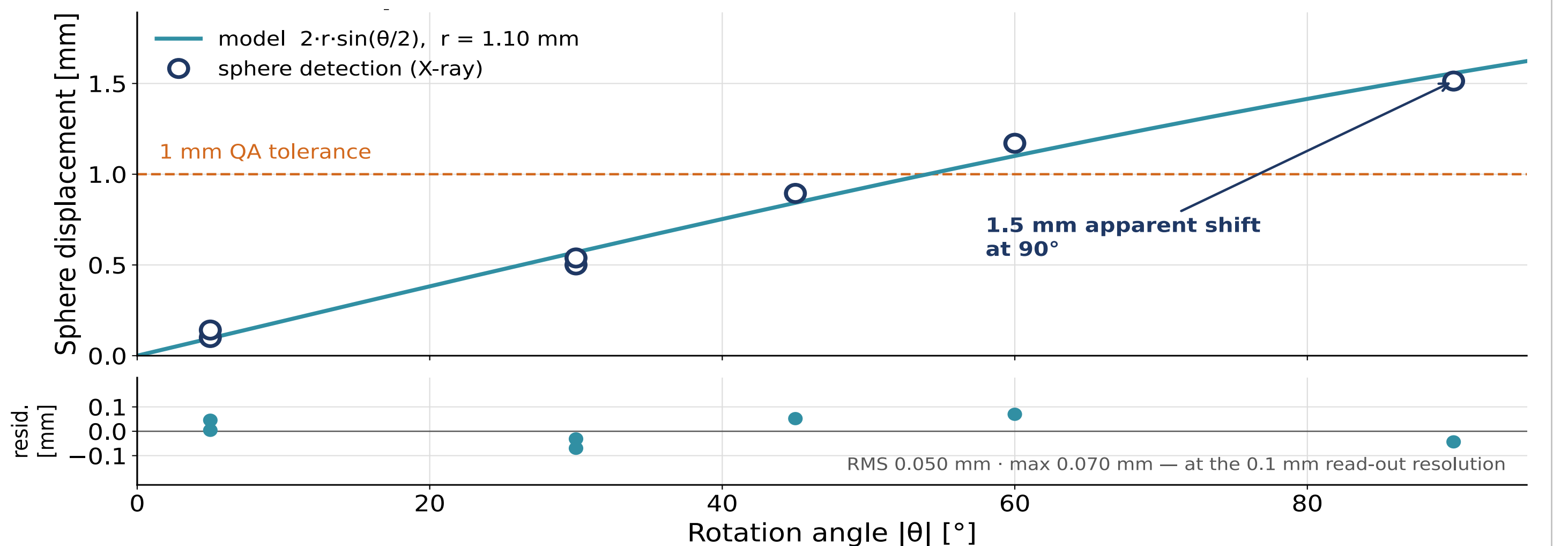


Fig. 8: The sphere sits off the rotation axis. Sphere-detection measurements at rotation angles from 5° to 90°, against the model for a sphere displaced by r from the rotation axis. The apparent shift reaches 1.5 mm at 90° phantom rotation, probably caused by print tolerances (ball bearing placement) or a misaligned bevel gear axis.

Discussion, Outlook & Code

Context. Table 4 of the ESTRO-ACROP guideline lists *D2 — Tracking performance* as a required test at 1 mm / 1° tolerance, monthly to annually on a C-arm linac. The document defines what has to be tested, how often, and to what tolerance — but the guideline gives no directive on how to achieve that number. That gap is the origin of this project. The same table drives the rest of it: *D3 respiratory trace*, *P1 thermal drift*, and *E1 End-to-End positioning* are the next entries the platform could be extended to cover. [1]

Next steps. In our construction the heated head is removable together with its platform, leaving a two-axis system. A breast-surface phantom on top of it moves on the trajectory shown above, bounded so it stays valid for dosimetry. From there: a gated versus ungated delivery compared on the EPID, closing the loop from tracking accuracy to delivered dose. Independent verification of the kinematic model by CT and a multi-point linearity run remain open.

Code. Current information about the mechatronic platform — firmware, serial protocol, terminal — and the QA evaluation with automated report generation are documented and available. Scan for the project page.



F. Final proof of function

- All axes driven ± 10 mm / $\pm 5^\circ$ (chronology: horizontal, vertical, rotational), aligned and evaluated — as-built and after calibration of the kinematic model over all Linacs via SD (see GitHub and QA documentation in QR code)

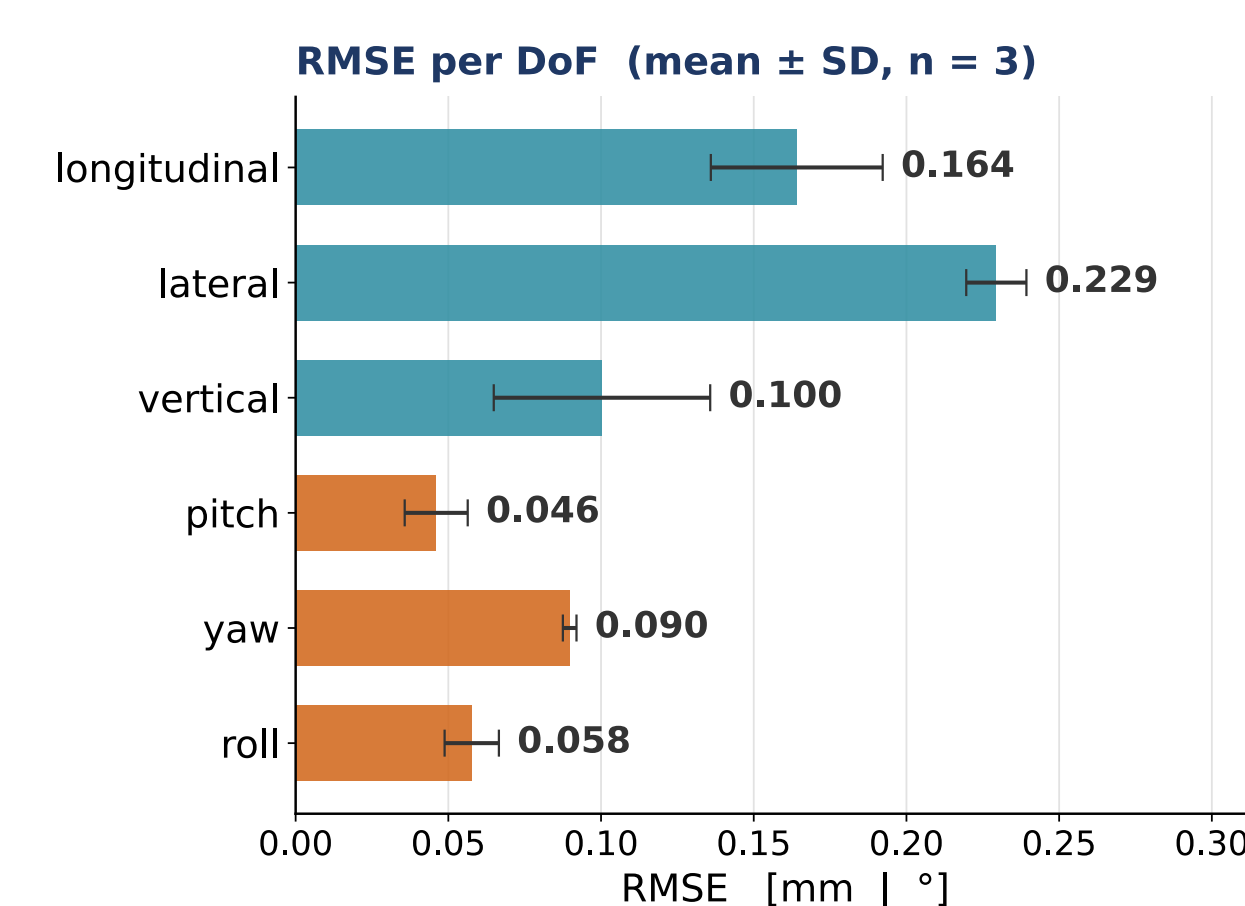
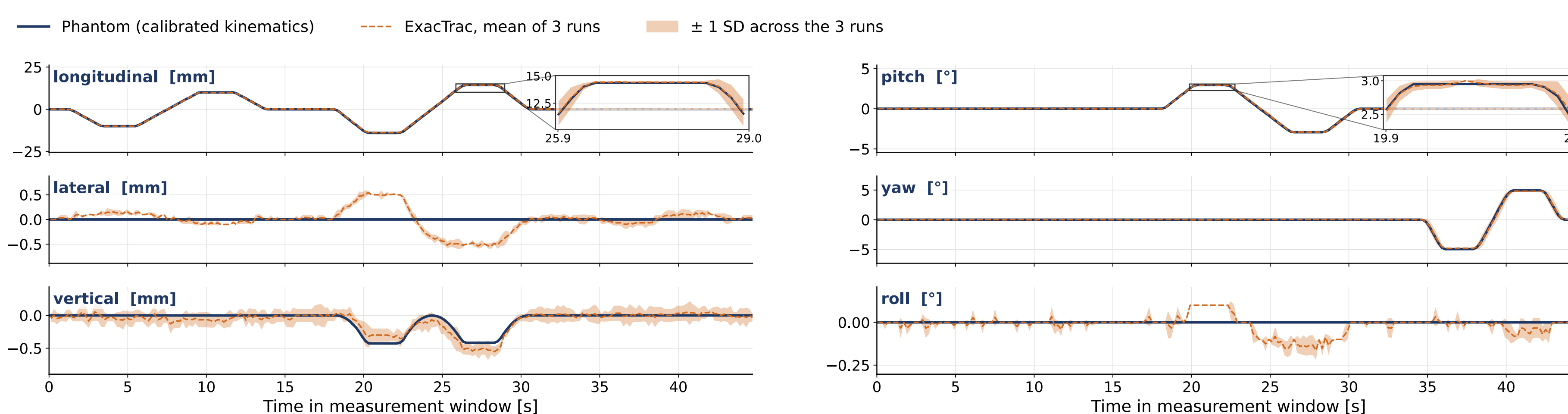


Fig. 9: Recorded data from three identical runs was aligned separately. The mean tracking values of all three runs were plotted against calculated phantom movement (left and middle panels). RMSE was calculated separately for each run and DoF, the mean and standard deviation is shown on the right. Tracked lateral movements of around 0.5 mm indicates slight misplacement of phantom or couch.